

Lab 3: SIFT Feature Extraction & Bag of Visual Words (BoVW) for Image Classification

Technical Report

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1. Dataset Overview

Dataset	UC Merced Land Use
Number of Classes	21 (e.g., airplane, river, harbor, freeway, buildings, forest, beach, etc.)
Images per Class	100
Total Images (after cleaning)	2,096
Image Size	256 × 256 pixels

2. Feature Extraction

SIFT (Scale-Invariant Feature Transform) descriptors were extracted from each image to capture local gradient and texture characteristics. Each descriptor is a 128-dimensional vector encoding distinctive visual patterns invariant to scale, rotation, and illumination changes. To represent each image globally, classical aggregation methods — namely the Mean, Max, and Min pooling of local descriptors — were applied. Furthermore, a Bag of Visual Words (BoVW) model was implemented by clustering all SIFT descriptors into visual vocabularies of various sizes ($k = 10, 50, 100, 150$), enabling the transformation of variable-length feature sets into fixed-length histogram representations suitable for classification.

3. Classification Setup

Two classifiers were tested to evaluate feature representations:

Classifier	Configuration
K-Nearest Neighbors (KNN)	$k = 5$
Support Vector Machine (SVM)	RBF Kernel, $C=1$, $\gamma='scale'$

Train/Test splits of 80/20 and 60/40 were used with stratified sampling to preserve class balance. The evaluation metric was **accuracy**.

4. Quantitative Results

Feature Type	KNN Accuracy	SVM Accuracy
Mean Descriptor	0.5670	0.6277
Max Descriptor	0.3556	0.4842
Min Descriptor	0.1793	0.1209
BoVW (k=10)	0.3967	0.4443
BoVW (k=50)	0.5384	0.6206
BoVW (k=100)	0.5420	0.6766
BoVW (k=150)	0.5628	0.6939

Observation: Across all feature representations, the SVM classifier consistently outperformed KNN, demonstrating its superior ability to model complex, non-linear decision boundaries. The best overall performance (69.39% accuracy) was achieved using the BoVW model with $k = 150$ visual words, highlighting the effectiveness of larger vocabularies in enhancing class separability.

5. Analysis of Results

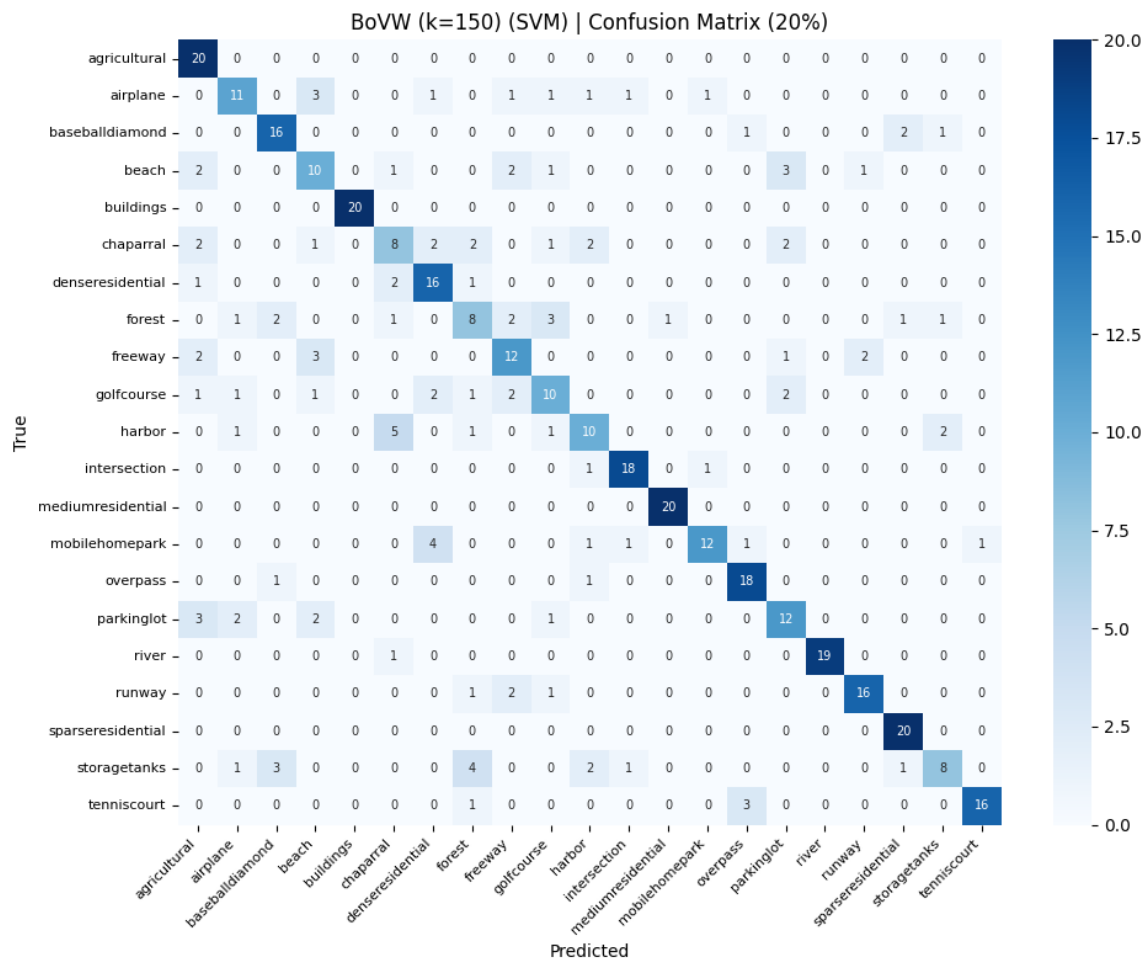
- **SVM vs. KNN:** The SVM classifier demonstrated superior performance, effectively capturing non-linear relationships in the feature space compared to KNN's distance-based approach.
- **Descriptor Stability:** Among classical descriptors, the Mean aggregation provided more stable and discriminative features than the Max or Min pooling strategies.
- **BoVW Effectiveness:** Expanding the visual vocabulary size improved class separability, with optimal performance achieved at $k = 150$, reflecting a richer representation of local patterns.
- **Overfitting Consideration:** Although larger vocabularies enhanced accuracy, they also increased the feature dimensionality, potentially leading to overfitting and higher computational cost.

6. Visual Evaluation

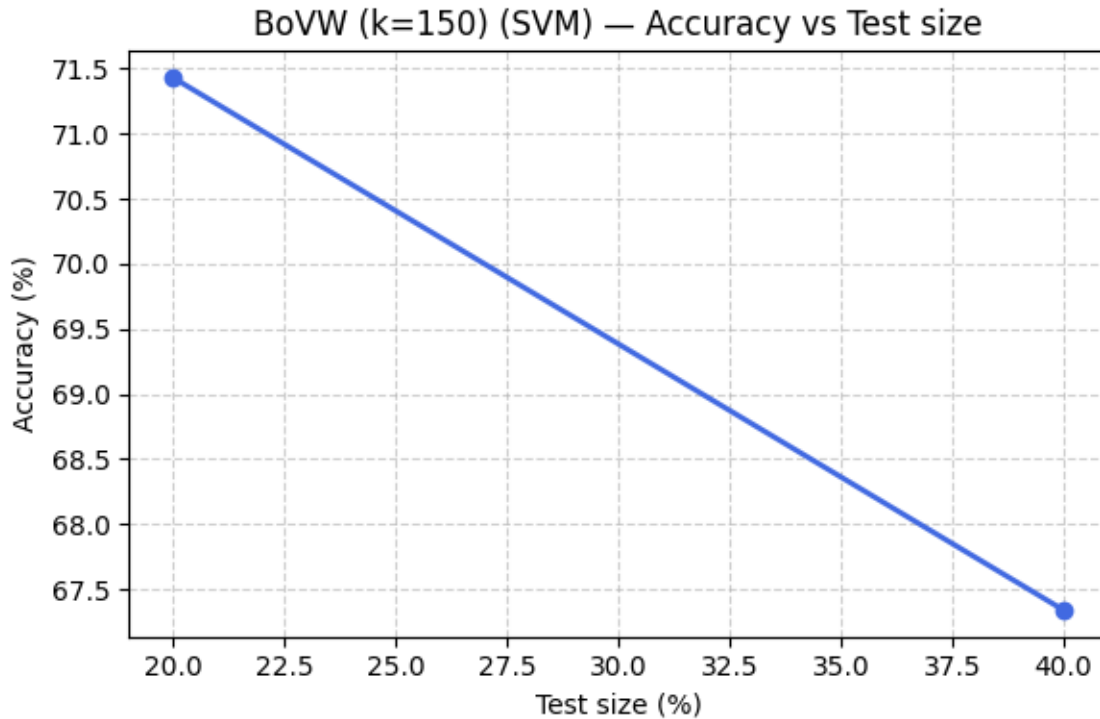
The confusion matrices indicated that visually similar land-use categories — such as *sparseresidential*, *mediumresidential*, and *denseresidential* — were frequently misclassified due to overlapping visual patterns and textures. In contrast, more distinct categories like *airplane*, *beach*, and *forest* exhibited significantly higher classification accuracy, reflecting the model's ability to differentiate between classes with clearly defined visual characteristics.

- The **best-performing model** was **BoVW (k = 150) + SVM**, achieving a 🏆 **71.43% accuracy** on the **20% test split**.
- The **confusion matrix** below illustrates strong diagonal dominance, confirming that the model correctly classified the majority of samples across 21 land-use categories.
- **High-performing categories:** *agricultural*, *buildings*, *mediumresidential*, *parkinglot*, and *river* achieved near-perfect classification accuracy.
- **Moderate confusion** was observed among visually similar categories such as *sparseresidential*, *mediumresidential*, and *denseresidential*, due to shared structural and textural characteristics.
- **Distinct categories** like *airplane*, *beach*, *forest*, and *golfcourse* maintained high recognition accuracy, reflecting the discriminative strength of the BoVW representation.
- The confusion matrix confirms **consistent and balanced performance**, with minimal bias across classes.

Final Confusion Matrix – BoVW (k = 150) + SVM (Test 20%)



Accuracy vs. Test Size Analysis – BoVW (k = 150) + SVM (Test 20%)



7. Conclusions

The **BoVW (Bag of Visual Words) combined with SVM** achieved the highest accuracy ($\approx 69\%$), demonstrating strong robustness and generalization for land-use image classification. In contrast, classical descriptors offered simpler yet less discriminative representations. These findings emphasize the effectiveness of **texture-based local feature aggregation** in capturing spatial and structural patterns, making BoVW a reliable and interpretable approach for **aerial scene classification**.